

Application Information

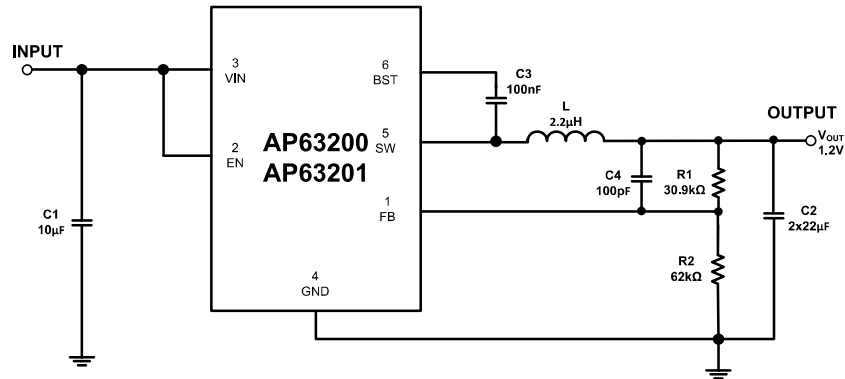


Figure 20. Typical Application Circuit of AP63200/AP63201

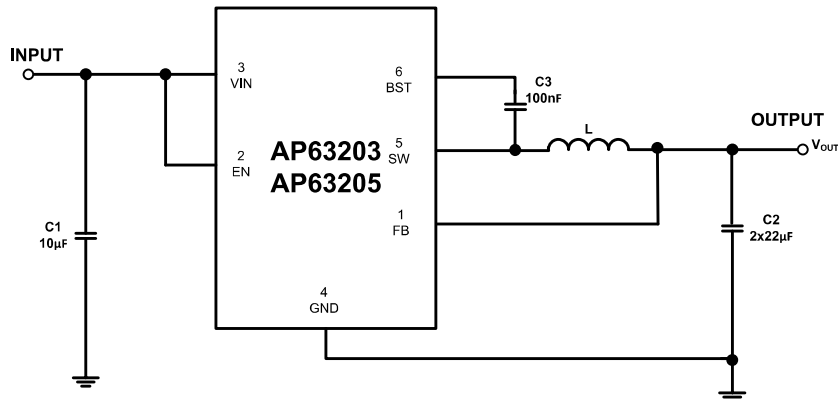


Figure 21. Typical Application Circuit of AP63203/AP63205

1 PWM Operation Control

The AP63200/AP63201/AP63203/AP63205 device is a 3.8V-to-32V input, 2A output, EMI friendly, fully integrated synchronous buck converter. Refer to the block diagram in Figure 3. The device employs fixed-frequency peak current mode control. The internal clock's rising edge (500kHz for the AP6300 and AP63201, 1.1MHz for the AP63203 and AP63205) initiates turning on the integrated high-side power MOSFET, Q1, for each cycle. When Q1 is on, the inductor current rises linearly, and the device charges the output capacitor. The current across Q1 is sensed and converted to a voltage with a ratio of R_T via the CSA block. The CSA output is combined with an internal slope compensation, S_E , resulting in V_{SUM} . When V_{SUM} rises higher than the internal COMP node, the device turns off Q1 and turns on the low-side power MOSFET, Q2. The inductor current decreases when Q2 is on. On the rising edge of next clock cycle, Q2 turns off, and Q1 turns on. This sequence repeats every clock cycle.

The peak current mode control with the internal loop compensation network and built-in 4ms soft-start simplifies the AP63200/AP63201/AP63203/AP63205 footprint as well as minimizes the external component count.

The error amplifier generates the COMP voltage by comparing the voltage on the FB pin with an internal 0.8V reference. An increase in load current causes the feedback voltage to drop. The error amplifier thus raises the COMP voltage until the average inductor current matches the increased load current. This feedback loop regulates the output voltage. The device also integrates internal slope compensation circuitry to prevent subharmonic oscillation when the duty cycle is greater than 50% for peak current mode control.

The AP63200/AP63201/AP63203/AP63205 device implements Frequency Spread Spectrum (FSS) with a switching frequency jitter of $\pm 6\%$. FSS reduces EMI by not allowing emitted energy to stay in any one frequency for a significant period of time. The converter further dampens high frequency radiated EMI noise through the use of its proprietary gate driver scheme to achieve a ringing-free switching node voltage without sacrificing the MOSFET switching times.

Layout

PCB Layout

1. The AP63200/AP63201/AP63203/AP63205 device works at 2A current load, so heat dissipation is a major concern in the layout of the PCB. 2oz copper for both the top and bottom layers is recommended.
2. Provide sufficient vias for the input and output capacitors' GND side to dissipate heat to the bottom layer.
3. Make the bottom layer under the device as the GND layer for heat dissipation. The GND layer should be as large as possible to provide better thermal effect.
4. Place the VIN capacitors as close to the device as possible.
5. Place the feedback components as close to FB as possible.
6. See Figure 25 for reference.

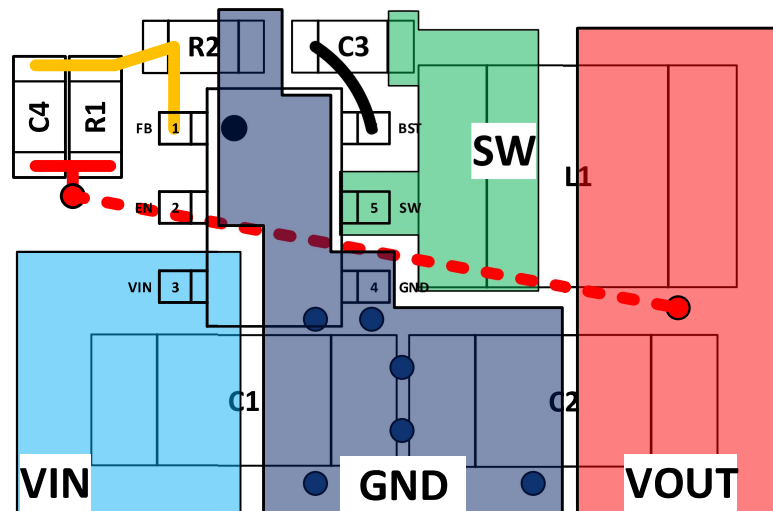


Figure 25. Recommended Layout